

How does the water that cools AI data centers 'use up' the water?

Posted by u/billybobamerica • • 0 points • 9 comments

I do not mean by using local water supplies, that one is an easy given in that it wastes water and steals it from residents.

I am more so wondering about the ones that are set up at the edge of bodies of water and cycle the water, or how the water is not returned to the system if it isn't. My understanding is that the water is running through the center and pulling heat away from the servers and components, but I fail to see how that is affecting overall water supply as its not being heated to the point of evaporation(total i mean, there's always some loss due to that but would it not just enter the water cycle at that low of a rate?) and should be returned to the body of water. In the case of locations not near bodies of water, could they not keep a closed loop with a decent sized reservoir to avoid waste? If you have useful reading suggestions throw them my way please.

Comments

u/DrToonhattan • 1 points • 2025-10-09 00:11 UTC

>its not being heated to the point of evaporation

Actually, in many cases it is. These data centers output so much heat that it's cheaper to use evaporative open loop cooling than have some system to cool the water down again as a closed loop would require.

u/HugePersonality1269 • 2 points • 2025-10-08 22:26 UTC

If they use Evaporative cooling towers you get small foot print and large cooling capacity. For every pound of water evaporated you get 978 BTU of cooling.... However this is a dog with fleas.

You lose water to evaporation. You lose water to drift (water droplets become airborne in the tower). As water evaporates dissolved solids concentrate up. This is reflected in a conductivity reading. Systems "blow down " or discharge tower water to drain - resulting fresh makeup water lowers conductivity.

The alternative is a large air to air cooler system. Lower water makeup requirements. Larger footprint.

More energy consumption in utilizing large fans to circulate air through the radiator coils. Also probably has a downside of more noise pollution.

u/Fearlessleader85 • 1 points • 2025-10-08 22:25 UTC

Water starts evaporating LONG before it boils. This is why sweating works. The water is lost through evaporation. And it's not GONE, it just joins the standard rain cycle through a nonstandard path. The potential problem is if it takes water from other paths to the point that is detrimental to them, or worse, if it pulls water from an aquifer faster than it can be refilled.

And this is not a problem everywhere. There are LOADS of places that have plenty of water and there's no concern. Also, this water doesn't need to be potable. It just needs to be clean enough and properly treated to not scale the heat exchangers.

People talking about how everywhere is going to run out of fresh water by 2035 or something are abusively simplifying a complex issue to the point of just being flat wrong. The next drainage over from where i live is running their aquifers dry. They're having to dig wells deeper and deeper and as the water table drops, the soil/rock collapses, causing loads of issues. It's a serious problem NOW, not in 10+ years. But my drainage is fine. We have all the water we need. We might only get 12-14" of precipitation per year, but the mountains above us get a lot more in snow and we get fed water from snowmelt until late september or October. It's a very localized problem.

u/left4ched • 1 points • 2025-10-08 22:16 UTC

I can only speak about closed loop cooling systems, and even though they're "closed" you still lose water because no equipment is perfect. Maintenance, repair, errors, broken parts, etc. Coolant is lost in a million teeny tiny ways that add up more the bigger the operation is.

u/Unlikely_Ad_7030 • 1 points • 2025-10-08 22:12 UTC

Quick Google search says yes, it is being heated up to the point of evaporation. Or at least one of the cooling methods used is called evaporative cooling.

Most of the water does go back into the reservoir or wherever they're getting it from, but a whole bunch of it is evaporated into the air as part of the cooling process. That air is then kinda just sent out of the building because hot humid air isn't gonna be any good for cooling purposes anymore, although I couldn't find much on this.

So when they however many gallons of water gets used up, it seems like yeah it's that many gallons of water getting turned into a gas.

Hopefully an actual expert answers tho lol

u/crashorbit • 3 points • 2025-10-08 22:07 UTC

Evaporation cooling towers are used on the hot side of data center AC loop.

u/generic_redditor_71 • 21 points • 2025-10-08 22:05 UTC

Many data centers and other industrial facilities that need to shed a lot of heat use wet cooling towers which evaporate some of the water going through them to cool the rest. The evaporated water isn't lost forever of course, it'll fall down as rain eventually somewhere, but it is effectively lost to the local area.

u/Conscious-Demand-594 • 3 points • 2025-10-08 22:04 UTC

There are two types of systems, evaporative and closed loop. The evaporative system loses water to the atmosphere through evaporation. The closed loop system uses active cooling which is more

expensive.

u/SLOBeachBoi • 19 points • 2025-10-08 22:04 UTC

Most of the big ones use evaporation cooling